

An Introduction to DAHA Involution and Macdonald Duality

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Main Statements

Statement (Root-system and DAHA involution). *For the dual affine root system R^t ,*

$$\alpha_j^t = \frac{A_j}{\sqrt{p}}, \quad \check{Q}^t = \frac{1}{\sqrt{p}}M, \quad M^t = \sqrt{p} \check{Q}. \quad (1)$$

There is an involution

$$\phi_{\tilde{W}} : \mathcal{H}_{\tilde{W}} \longrightarrow \mathcal{H}_{\tilde{W}^t} \quad (2)$$

given by

$$\phi_{\tilde{W}}(T_j) = (T_j^t)^{-1}, \quad \phi_{\tilde{W}}(Y_\mu) = X_{\mu/\sqrt{p}}^t, \quad \phi_{\tilde{W}}(X_\beta) = Y_{\sqrt{p}\beta}^t, \quad (3)$$

and

$$\phi_{\tilde{W}}(q) = (q^t)^{-1}, \quad \phi_{\tilde{W}}(t_j^{1/2}) = (t_j^t)^{-1/2}. \quad (4)$$

For $B_n^{(1)}$, we have $(B_n^{(1)})^t = C_n^{(1)}$ and $p = 2$.

Statement (Non-symmetric Macdonald polynomials). *For generic parameters and $\lambda \in \check{Q}$, there is a unique monic triangular common Y -eigenvector*

$$E_\lambda^R = X_\lambda + \sum_{\eta < \lambda} c_{\lambda\eta}^R X_\eta, \quad Y_\mu E_\lambda^R = \gamma_\lambda^R(\mu) E_\lambda^R \quad (\mu \in M). \quad (5)$$

Statement (Non-symmetric evaluation-duality). *Define*

$$X_{\mu/\sqrt{p}}^t(\text{sp}_R(\lambda)) = \gamma_\lambda^R(-\mu), \quad X_\lambda(\text{sp}_{R^t}(\beta)) = \gamma_\beta^{R^t}(-\sqrt{p}\lambda). \quad (6)$$

Then

$$\boxed{\frac{E_\lambda^R(\text{sp}_{R^t}(\beta))}{E_\lambda^R(\text{sp}_{R^t}(0))} = \frac{E_\beta^{R^t}(\text{sp}_R(\lambda))}{E_\beta^{R^t}(\text{sp}_R(0))}}. \quad (7)$$

The details and the $B_n^{(1)}-C_n^{(1)}$ calculation follow.

Contents

1	Notation and Conventions	2
2	Root Data	2
3	The Double Affine Hecke Algebra	4
3.1	The finite Hecke algebra	4
3.2	The affine Hecke algebra	5
3.3	The double affine Hecke algebra	5
4	The Root-System and DAHA Involutions	5
4.1	The root-system involution	5
4.2	The DAHA isomorphism	6
5	Non-symmetric Macdonald Polynomials	6
6	Macdonald Duality	8
6.1	Spectral points	8
6.2	The duality theorem	8
7	Examples	8
7.1	Parameter conventions for the B_n – C_n calculation	8
7.2	The root-system involution	9
7.3	The polynomial $E_{\varepsilon_1}^{B_n^{(1)}}$	10
7.4	The corresponding polynomial of type $C_n^{(1)}$	13
7.5	An explicit B_n – C_n duality identity	15
8	Guiding Principle	16

1 Notation and Conventions

We use Ion's conventions. Thus $X_\delta = q^{-1}$, the X -lattice is $\mathring{Q}$, and the Y -lattice is M . Operators act from right to left.

Symbol	Meaning
R, R^ι	An affine root system and its dual.
$\alpha_0, \dots, \alpha_n$	The affine simple roots; $\alpha_1, \dots, \alpha_n$ are the finite simple roots.
δ, θ	The null root and the highest finite root; $\alpha_0 = \delta - \theta$.
$\mathring{Q}$	The finite root lattice and the index lattice for X .
M	The translation lattice generated by $A_j = e_j \alpha_j$, and the index lattice for Y .
$\mathring{W}, W$	The finite and affine Weyl groups.
λ_μ	Translation by $\mu \in M$.
X_β, Y_μ	The two commuting lattice directions in DAHA.
T_j	The Hecke generator for s_j .
$p = e_s$	The scaling number; $p = 2$ in types $B_n^{(1)}$ and $C_n^{(1)}$.
$\phi_{\tilde{W}}$	The DAHA Involution.
$\gamma_\lambda^R(\mu)$	The Y_μ -eigenvalue of E_λ^R .
$\text{sp}_R(\lambda)$	The spectral character attached to λ .
E_λ^R	The non-symmetric Macdonald polynomial.

2 Root Data

Let

$$A = (a_{ij})_{0 \leq i, j \leq n} \quad (8)$$

be an irreducible affine Cartan matrix, with finite part

$$\mathring{A} = (a_{ij})_{1 \leq i, j \leq n}. \quad (9)$$

Let

$$\alpha_0, \alpha_1, \dots, \alpha_n \quad (10)$$

be the affine simple roots, and set

$$\mathring{Q} = \bigoplus_{j=1}^n \mathbb{Z} \alpha_j. \quad (11)$$

Write

$$\delta = \sum_{j=0}^n a_j \alpha_j, \quad \delta^\vee = \sum_{j=0}^n a_j^\vee \alpha_j^\vee. \quad (12)$$

We exclude type $A_{2n}^{(2)}$. Then $a_0 = 1$ and

$$\alpha_0 = \delta - \theta, \quad (13)$$

where θ is the highest finite root.

For $1 \leq j \leq n$, set

$$d_j := \frac{a_j}{a_j^\vee}, \quad e_j := \max\{d_j, a_0^{-1}\}, \quad A_j := e_j \alpha_j, \quad (14)$$

and

$$M := \bigoplus_{j=1}^n \mathbb{Z} A_j. \quad (15)$$

Write λ_μ for translation by $\mu \in M$. The value of e_j depends only on the root length. Let e_s and e_l be its values on short and long roots, and set

$$p := e_s. \quad (16)$$

In the simply laced case, $p = 1$.

Use the symmetric form

$$(\alpha_j, \alpha_k) := d_j^{-1} a_{jk}, \quad 0 \leq j, k \leq n, \quad (17)$$

and the reflections

$$s_\alpha(x) = x - \frac{2(x, \alpha)}{(\alpha, \alpha)} \alpha. \quad (18)$$

The finite Weyl group has the presentation

$$\mathring{W} = \left\langle s_i \ (1 \leq i \leq n) \ \middle| \ s_i^2 = 1, \ \underbrace{s_i s_j s_i \cdots}_{m_{ij} \text{ factors}} = \underbrace{s_j s_i s_j \cdots}_{m_{ij} \text{ factors}} \ (i \neq j) \right\rangle, \quad (19)$$

where

$$m_{ij} = \begin{cases} 2, & a_{ij} a_{ji} = 0, \\ 3, & a_{ij} a_{ji} = 1, \\ 4, & a_{ij} a_{ji} = 2, \\ 6, & a_{ij} a_{ji} = 3, \\ \infty, & a_{ij} a_{ji} \geq 4. \end{cases} \quad (20)$$

See [Kac83, §3.13]. The affine Weyl group is

$$W = \mathring{W} \ltimes M. \quad (21)$$

3 The Double Affine Hecke Algebra

Let m be the least common denominator of

$$\{(\alpha_j, \Lambda_k) \mid 1 \leq j, k \leq n\}, \quad (22)$$

where Λ_k is the k -th fundamental weight. Let $\mathbb{F}$ be the field of rational functions in $q^{1/m}$ and the $t_j^{1/2}$.

The two lattice algebras are

$$\mathbb{F}[M] = \bigoplus_{\mu \in M} \mathbb{F}Y_\mu, \quad \mathbb{F}[\dot{Q}] = \bigoplus_{\beta \in \dot{Q}} \mathbb{F}X_\beta, \quad (23)$$

with

$$Y_\mu Y_\nu = Y_{\mu+\nu}, \quad X_\beta X_\gamma = X_{\beta+\gamma}. \quad (24)$$

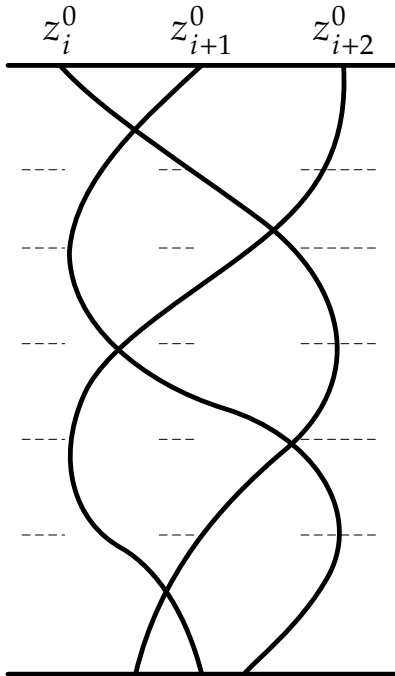
The parameters depend only on root length:

$$d_j = d_k \iff t_j = t_k. \quad (25)$$

3.1 The finite Hecke algebra

Since we omitted the reflection relations one may want to know what is the inverse of T_j . The finite Hecke algebra $\mathcal{H}_{\dot{W}}$ is generated by $T_1, \dots, T_n$. It has the braid relations from (19) and the quadratic relations

$$T_j - T_j^{-1} = t_j^{1/2} - t_j^{-1/2}. \quad (26)$$



$$(c) \quad \sigma_i \sigma_{i+1} \sigma_i \sigma_{i+1}^{-1} \sigma_i^{-1} \sigma_{i+1}^{-1} = \text{identity}$$

3.2 The affine Hecke algebra

The affine Hecke algebra $\mathcal{H}_W$ is generated by $\mathcal{H}_{\tilde{W}}$ and $\mathbb{F}[M]$, with

$$Y_\mu T_j - T_j Y_{s_j(\mu)} = \left(t_j^{1/2} - t_j^{-1/2} \right) \frac{Y_\mu - Y_{s_j(\mu)}}{1 - Y_{A_j}}, \quad (27)$$

for $\mu \in M$ and $1 \leq j \leq n$.

3.3 The double affine Hecke algebra

Definition 3.1. The double affine Hecke algebra $\mathcal{H}_{\tilde{W}}$ is generated by $\mathcal{H}_W$ and $\mathbb{F}[\mathring{Q}]$, with

$$T_j X_\beta - X_{s_j(\beta)} T_j = \left(t_j^{1/2} - t_j^{-1/2} \right) \frac{X_\beta - X_{s_j(\beta)}}{1 - X_{-\alpha_j}}, \quad (28)$$

for $\beta \in \mathring{Q}$ and $0 \leq j \leq n$, together with

$$X_\delta = q^{-1}. \quad (29)$$

For $j = 0$, use $X_\delta = q^{-1}$ to read $X_{s_0(\beta)}$.

Thus DAHA contains the two commuting lattice algebras

$$\mathbb{F}[\mathring{Q}] \quad \text{generated by the } X_\beta, \quad (30)$$

and

$$\mathbb{F}[M] \quad \text{generated by the } Y_\mu. \quad (31)$$

4 The Root-System and DAHA Involutions

4.1 The root-system involution

Definition 4.1. For $1 \leq j \leq n$, define

$$\alpha_j^\iota := \frac{A_j}{\sqrt{p}} = \frac{e_j}{\sqrt{p}} \alpha_j, \quad (32)$$

and set

$$\delta^\iota = \delta. \quad (33)$$

These data define the affine root system R^ι . Its affine simple root α_0^ι is fixed by its affine type.

Under the natural identification of the finite root spaces,

$$\mathring{Q}^\iota = \frac{1}{\sqrt{p}} M, \quad M^\iota = \sqrt{p} \mathring{Q}. \quad (34)$$

The simply laced types are fixed. In particular,

$$A_n^{(1)\iota} = A_n^{(1)}. \quad (35)$$

For the classical non-simply-laced types,

$$B_n^{(1)\iota} = C_n^{(1)}, \quad C_n^{(1)\iota} = B_n^{(1)}. \quad (36)$$

The map ι swaps short and long finite roots.

4.2 The DAHA isomorphism

Write the generators of $\mathcal{H}_{\tilde{W}^t}$ as

$$T_j^t, \quad X_\gamma^t, \quad Y_v^t. \quad (37)$$

Theorem 4.2 (Ion's DAHA involution). *The assignments*

$$\phi_{\tilde{W}}(T_j) = (T_j^t)^{-1}, \quad 1 \leq j \leq n, \quad (38)$$

$$\phi_{\tilde{W}}(Y_\mu) = X_{\mu/\sqrt{p}}^t, \quad \mu \in M, \quad (39)$$

$$\phi_{\tilde{W}}(X_\beta) = Y_{\sqrt{p}\beta}^t, \quad \beta \in \mathring{Q}, \quad (40)$$

extend to an involution

$$\phi_{\tilde{W}} : \mathcal{H}_{\tilde{W}} \longrightarrow \mathcal{H}_{\tilde{W}^t}, \quad (41)$$

with

$$\phi_{\tilde{W}}(q) = (q^t)^{-1}, \quad \phi_{\tilde{W}}(t_j^{1/2}) = (t_j^t)^{-1/2}. \quad (42)$$

Its inverse is $\phi_{\tilde{W}^t}$.

This is [Ion03, Theorems 2.2–2.3]. In short,

$$\boxed{X_\beta \longleftrightarrow Y_{\sqrt{p}\beta}^t}. \quad (43)$$

5 Non-symmetric Macdonald Polynomials

The polynomial representation is

$$\mathcal{P}_R := \mathbb{F}[\mathring{Q}]. \quad (44)$$

For $\lambda \in \mathring{Q}$, let λ_+ be the dominant element of $\mathring{W}\lambda$. Let $v(\lambda)$ be the shortest element of $\mathring{W}$ that sends λ to the anti-dominant chamber. Define

$$\lambda \geq \eta \iff \begin{cases} \lambda_+ > \eta_+, & \text{in strict dominance order,} \\ \lambda_+ = \eta_+ \text{ and } v(\lambda) \leq v(\eta), & \text{in finite Bruhat order.} \end{cases} \quad (45)$$

For dominant elements $\xi, \zeta \in \mathring{Q}$, the dominance order is defined by

$$\xi \leq_{\text{dom}} \zeta \iff \xi - \zeta \in \sum_{i=1}^n \mathbb{Z}_{\geq 0} \alpha_i. \quad (46)$$

Set

$$\mathring{Q}_{\leq \lambda} := \{\eta \in \mathring{Q} \mid \eta \leq \lambda\}, \quad \mathcal{P}_{\leq \lambda} := \text{span}_{\mathbb{F}}\{X_\eta \mid \eta \leq \lambda\}. \quad (47)$$

The set $\mathring{Q}_{\leq \lambda}$ is finite and lower closed.

Theorem 5.1 (Y -triangularity). For $\mu \in M$ and $\lambda \in \mathring{Q}$, there are unique coefficients $\gamma_\lambda^R(\mu) \in \mathbb{F}^\times$ and $a_{\eta\lambda}^R(\mu) \in \mathbb{F}$ such that

$$Y_\mu X_\lambda = \gamma_\lambda^R(\mu) X_\lambda + \sum_{\eta < \lambda} a_{\eta\lambda}^R(\mu) X_\eta. \quad (48)$$

Hence every Y_μ preserves $\mathcal{P}_{\leq \lambda}$, see [Mac03, 4.6.4–4.6.12].

Definition 5.2. For generic parameters, $E_\lambda^R \in \mathcal{P}_R$ is the monic common eigenvector defined by

$$E_\lambda^R = X_\lambda + \sum_{\eta < \lambda} c_{\lambda\eta}^R X_\eta, \quad (49)$$

$$Y_\mu E_\lambda^R = \gamma_\lambda^R(\mu) E_\lambda^R, \quad \mu \in M. \quad (50)$$

Here generic means that the required eigenvalues are distinct and the denominators below are nonzero.

Proposition 5.3 (Triangular recursion). Choose $\nu \in M$ such that the eigenvalues $\gamma_\eta^R(\nu)$, $\eta \leq \lambda$, are pairwise distinct.

Fix $\eta < \lambda$. Suppose that all coefficients $c_{\lambda\zeta}^R$ with $\eta < \zeta < \lambda$ are already known. Define

$$E_\lambda^{(>\eta)} := X_\lambda + \sum_{\eta < \zeta < \lambda} c_{\lambda\zeta}^R X_\zeta. \quad (51)$$

This is the part of E_λ^R already computed above η . It is only a partial sum, not a new Macdonald polynomial.

Then

$$c_{\lambda\eta}^R = \frac{[X_\eta] Y_\nu E_\lambda^{(>\eta)}}{\gamma_\lambda^R(\nu) - \gamma_\eta^R(\nu)}. \quad (52)$$

Thus the coefficients are computed from top to bottom.

Example 5.4. Suppose

$$\lambda > \zeta > \eta \quad (53)$$

and

$$Y_\nu X_\lambda = \gamma_\lambda X_\lambda + a_{\zeta\lambda} X_\zeta + a_{\eta\lambda} X_\eta + \cdots, \quad (54)$$

$$Y_\nu X_\zeta = \gamma_\zeta X_\zeta + a_{\eta\zeta} X_\eta + \cdots. \quad (55)$$

First,

$$c_{\lambda\zeta} = \frac{a_{\zeta\lambda}}{\gamma_\lambda - \gamma_\zeta}. \quad (56)$$

When computing $c_{\lambda\eta}$, the partial sum is

$$E_\lambda^{(>\eta)} = X_\lambda + c_{\lambda\zeta} X_\zeta. \quad (57)$$

Therefore

$$[X_\eta] Y_\nu E_\lambda^{(>\eta)} = a_{\eta\lambda} + c_{\lambda\zeta} a_{\eta\zeta}, \quad (58)$$

and hence

$$c_{\lambda\eta} = \frac{a_{\eta\lambda} + c_{\lambda\zeta} a_{\eta\zeta}}{\gamma_\lambda - \gamma_\eta}. \quad (59)$$

Since $T_i 1 = t_i^{1/2} 1$, every Y_μ acts on 1 by a scalar. Therefore

$$\boxed{E_0^R = 1.} \quad (60)$$

6 Macdonald Duality

6.1 Spectral points

For $\lambda \in \mathring{Q}$, define the character

$$\mathrm{sp}_R(\lambda) : \mathcal{P}_{R^\iota} \longrightarrow \mathbb{F}, \quad \boxed{X_{\mu/\sqrt{p}}^\iota(\mathrm{sp}_R(\lambda)) := \gamma_\lambda^R(-\mu)}, \quad \mu \in M. \quad (61)$$

This defines it on all of $\mathcal{P}_{R^\iota} = \mathbb{F}[\mathring{Q}^\iota]$, since $\mathring{Q}^\iota = p^{-1/2}M$. For $\beta \in \mathring{Q}^\iota$, define

$$\mathrm{sp}_{R^\iota}(\beta) : \mathcal{P}_R \longrightarrow \mathbb{F}, \quad \boxed{X_\lambda(\mathrm{sp}_{R^\iota}(\beta)) := \gamma_\beta^{R^\iota}(-\sqrt{p}\lambda)}, \quad \lambda \in \mathring{Q}. \quad (62)$$

6.2 The duality theorem

Theorem 6.1. [*Mac03*, 4.7.14] For $\lambda \in \mathring{Q}$ and $\beta \in \mathring{Q}^\iota$,

$$\boxed{\frac{E_\lambda^R(\mathrm{sp}_{R^\iota}(\beta))}{E_\lambda^R(\mathrm{sp}_{R^\iota}(0))} = \frac{E_\beta^{R^\iota}(\mathrm{sp}_R(\lambda))}{E_\beta^{R^\iota}(\mathrm{sp}_R(0))}.} \quad (63)$$

7 Examples

7.1 Parameter conventions for the B_n – C_n calculation

Write

$$q_B, \quad t_l^B, \quad t_s^B \quad (64)$$

for the parameters of $B_n^{(1)}$, and

$$q_C, \quad t_s^C, \quad t_l^C \quad (65)$$

for those of $C_n^{(1)}$. The subscripts always refer to root length in the indicated root system.

Remark 7.1 (Parameter convention in duality). *The map ϕ_{BC} produces the $C_n^{(1)}$ -realization with inverted parameters*

$$\phi_{BC}(q_B) = q_C^{-1}, \quad \phi_{BC}(t_l^B) = (t_s^C)^{-1}, \quad \phi_{BC}(t_s^B) = (t_l^C)^{-1}, \quad (66)$$

whereas $E_\beta^{C_n^{(1)}}$ is written in the standard $C_n^{(1)}$ -parameters. Thus we need to introduce

$$\overline{q_C} = q_C^{-1}, \quad \overline{t_s^C} = (t_s^C)^{-1}, \quad \overline{t_l^C} = (t_l^C)^{-1}. \quad (67)$$

Then the coefficient identification used in evaluation-duality is

$$j_{BC} := {}^- \circ \phi_{BC}, \quad (68)$$

so that

$$j_{BC}(q_B) = q_C, \quad j_{BC}(t_l^B) = t_s^C, \quad j_{BC}(t_s^B) = t_l^C. \quad (69)$$

7.2 The root-system involution

The finite simple roots of types A , B , and C are

$$\begin{aligned} A_{n-1} : \quad & \alpha_i = \varepsilon_i - \varepsilon_{i+1}, & 1 \leq i < n; \\ B_n : \quad & \alpha_i = \varepsilon_i - \varepsilon_{i+1}, & 1 \leq i < n, & \alpha_n = \varepsilon_n; \\ C_n : \quad & \alpha_i = \varepsilon_i - \varepsilon_{i+1}, & 1 \leq i < n, & \alpha_n = 2\varepsilon_n. \end{aligned}$$

Table 1: Affine labels and dual labels.

Type	$(a_0, a_1, \dots, a_n)$	$(a_0^\vee, a_1^\vee, \dots, a_n^\vee)$
$A_n^{(1)}$	$(1, 1, \dots, 1)$	$(1, 1, \dots, 1)$
$B_n^{(1)}$	$(1, 1, 2, \dots, 2, 2)$	$(1, 1, 2, \dots, 2, 1)$
$C_n^{(1)}$	$(1, 2, \dots, 2, 1)$	$(1, 1, \dots, 1)$

Hence

$$A_n^{(1)} : \quad M = \mathring{Q}, \quad (70)$$

$$B_n^{(1)} : \quad M = \bigoplus_{i=1}^{n-1} \mathbb{Z}\alpha_i \oplus \mathbb{Z}(2\alpha_n), \quad (71)$$

$$C_n^{(1)} : \quad M = \bigoplus_{i=1}^{n-1} \mathbb{Z}(2\alpha_i) \oplus \mathbb{Z}\alpha_n. \quad (72)$$

Now take $B_n^{(1)}$. Then

$$d_1 = \dots = d_{n-1} = 1, \quad d_n = 2, \quad (73)$$

so

$$e_1 = \dots = e_{n-1} = 1, \quad e_n = 2. \quad (74)$$

Thus

$$A_j = \alpha_j \quad (1 \leq j < n), \quad A_n = 2\alpha_n, \quad p = e_s = 2. \quad (75)$$

Therefore

$$\alpha_i^t = \frac{\varepsilon_i - \varepsilon_{i+1}}{\sqrt{2}} \quad (i < n), \quad \alpha_n^t = \sqrt{2} \varepsilon_n. \quad (76)$$

Their squared lengths are

$$(\alpha_i^t, \alpha_i^t) = 1 \quad (i < n), \quad (\alpha_n^t, \alpha_n^t) = 2. \quad (77)$$

The first $n - 1$ roots are short and the last is long. The finite Cartan matrix is

$$\mathring{A}^t = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & -1 & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & 2 & -2 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}, \quad (78)$$

so

$$\boxed{(B_n^{(1)})^t = C_n^{(1)}}. \quad (79)$$

In this case, Theorem 4.2 reads

$$\phi_{\tilde{W}}(T_j) = (T_j^t)^{-1}, \quad 1 \leq j \leq n, \quad (80)$$

$$\phi_{\tilde{W}}(Y_\mu) = X_{\mu/\sqrt{2}}^t, \quad \mu \in M, \quad (81)$$

$$\phi_{\tilde{W}}(X_\beta) = Y_{\sqrt{2}\beta}^t, \quad \beta \in \mathring{Q}, \quad (82)$$

$$\phi_{\tilde{W}}(q) = (q^t)^{-1}, \quad (83)$$

$$\phi_{\tilde{W}}(t_j) = (t_j^t)^{-1}. \quad (84)$$

Also,

$$\mathring{Q}^t = \frac{1}{\sqrt{2}}M = \bigoplus_{i=1}^{n-1} \mathbb{Z} \frac{\varepsilon_i - \varepsilon_{i+1}}{\sqrt{2}} \oplus \mathbb{Z}(\sqrt{2}\varepsilon_n), \quad (85)$$

and

$$M^t = \bigoplus_{i=1}^{n-1} \mathbb{Z}(2\alpha_i^t) \oplus \mathbb{Z}\alpha_n^t = \sqrt{2}\mathring{Q}. \quad (86)$$

Particularly, we have

$$\boxed{\phi_{\tilde{W}}\left((t_l^B)^{1/2}\right) = (t_s^C)^{-1/2}, \quad \phi_{\tilde{W}}\left((t_s^B)^{1/2}\right) = (t_l^C)^{-1/2}}. \quad (87)$$

Thus the X-lattice of B_n is the Y-lattice of C_n , and the Y-lattice of B_n is the X-lattice of C_n .

7.3 The polynomial $E_{\varepsilon_1}^{B_n^{(1)}}$

Lemma 7.2 (Triangular support of ε_1). *For the order in (45),*

$$\{\eta \in \mathring{Q} : \eta \leq \varepsilon_1\} = \{\varepsilon_1, 0\}. \quad (88)$$

Consequently,

$$\mathcal{P}_{\leq \varepsilon_1} = \text{span}_{\mathbb{F}}\{x_1, 1\}. \quad (89)$$

Proof. Write

$$\eta_+ = \sum_{j=1}^n m_j \varepsilon_j. \quad (90)$$

For type B_n ,

$$\alpha_i^\vee = \varepsilon_i - \varepsilon_{i+1} \quad (i < n), \quad \alpha_n^\vee = 2\varepsilon_n. \quad (91)$$

Hence

$$(\eta_+, \alpha_i^\vee) = m_i - m_{i+1} \quad (i < n), \quad (\eta_+, \alpha_n^\vee) = 2m_n. \quad (92)$$

Therefore η_+ is dominant if and only if

$$m_1 \geq m_2 \geq \cdots \geq m_n \geq 0. \quad (93)$$

Suppose that $\eta_+ \leq_{\text{dom}} \varepsilon_1$. Then

$$\varepsilon_1 - \eta_+ = \sum_{i=1}^n c_i \alpha_i, \quad c_i \geq 0. \quad (94)$$

Comparing coordinates gives

$$1 - m_1 = c_1, \quad -m_j = c_j - c_{j-1} \quad (2 \leq j \leq n). \quad (95)$$

If $\eta_+ \neq 0$, then $m_1 \geq 1$. Thus $m_1 = 1$ and $c_1 = 0$. The remaining equations, together with $m_j, c_j \geq 0$, give successively

$$m_j = c_j = 0 \quad (2 \leq j \leq n). \quad (96)$$

Hence η_+ is either 0 or ε_1 .

It remains to consider

$$\dot{W}_{\varepsilon_1} = \{\pm \varepsilon_i : 1 \leq i \leq n\}. \quad (97)$$

For the shortest element sending a weight to $-\varepsilon_1$,

$$\ell(v(\varepsilon_i)) = 2n - i, \quad \ell(v(-\varepsilon_i)) = i - 1. \quad (98)$$

Thus $v(\varepsilon_1)$ is the unique element of maximal length. Since Bruhat order is length-increasing,

$$v(\varepsilon_1) \leq v(\eta) \quad (99)$$

forces $\eta = \varepsilon_1$. □

It follows that

$$E_{\varepsilon_1}^{B_n^{(1)}} = x_1 + c_B. \quad (100)$$

The operators in this formula carry their $B_n^{(1)}$ -parameters. where

$$t_0^B = t_1^B = \cdots = t_{n-1}^B = t_l^B, \quad t_n^B = t_s^B. \quad (101)$$

For $1 \leq i < n$, s_i swaps x_i and x_{i+1} . The reflection s_n sends x_n to x_n^{-1} , and

$$(s_0 f)(x_1, x_2, x_3, \dots, x_n) = f\left(\frac{1}{qx_2}, \frac{1}{qx_1}, x_3, \dots, x_n\right). \quad (102)$$

Set

$$L_B := (t_l^B)^{1/2}, \quad S_B := (t_s^B)^{1/2}. \quad (103)$$

Since $X_{-\alpha_0} = qx_1x_2$,

$$T_i f = L_B s_i f + (L_B - L_B^{-1}) \frac{f - s_i f}{1 - x_{i+1}/x_i}, \quad 1 \leq i < n, \quad (104)$$

$$T_n f = S_B s_n f + (S_B - S_B^{-1}) \frac{f - s_n f}{1 - x_n^{-1}}, \quad (105)$$

$$T_0 f = L_B s_0 f + (L_B - L_B^{-1}) \frac{f - s_0 f}{1 - qx_1x_2}. \quad (106)$$

For $\theta = \varepsilon_1 + \varepsilon_2$, let

$$R_2^B = T_2 T_3 \cdots T_{n-1} T_n T_{n-1} \cdots T_3 T_2. \quad (107)$$

By Ion's result ,

$$Y_{-\theta}^{B_n^{(1)}} = R_2^B T_1 R_2^B T_0. \quad (108)$$

From

$$s_0 x_1 = q_B^{-1} x_2^{-1}, \quad \frac{x_1 - q_B^{-1} x_2^{-1}}{1 - q_B x_1 x_2} = -q_B^{-1} x_2^{-1}, \quad (109)$$

we obtain

$$T_0 x_1 = q_B^{-1} L_B^{-1} x_2^{-1}. \quad (110)$$

Applying (108) from right to left gives

$$\boxed{Y_{-\theta}^{B_n^{(1)}} x_1 = q_B^{-1} (t_l^B)^{-1} x_1 - q_B^{-1} (t_l^B)^{n-2} (t_s^B - 1).} \quad (111)$$

Since the word in (108) contains $4n-6$ long-root generators and two short-root generators,

$$\boxed{Y_{-\theta}^{B_n^{(1)}} 1 = (t_l^B)^{2n-3} t_s^B.} \quad (112)$$

Set

$$a_B := q_B^{-1} (t_l^B)^{-1}, \quad (113)$$

$$b_B := -q_B^{-1} (t_l^B)^{n-2} (t_s^B - 1), \quad (114)$$

$$d_B := (t_l^B)^{2n-3} t_s^B. \quad (115)$$

Then

$$\gamma_{\varepsilon_1}^{B_n^{(1)}}(-\theta) = a_B, \quad \gamma_0^{B_n^{(1)}}(-\theta) = d_B. \quad (116)$$

If $a_B \neq d_B$, the eigenvalue equation for (100) gives

$$c_B = \frac{q_B^{-1}(t_l^B)^{n-2}(t_s^B - 1)}{(t_l^B)^{2n-3}t_s^B - q_B^{-1}(t_l^B)^{-1}}. \quad (117)$$

Therefore

$$E_{\varepsilon_1}^{B_n^{(1)}} = x_1 + \frac{q_B^{-1}(t_l^B)^{n-2}(t_s^B - 1)}{(t_l^B)^{2n-3}t_s^B - q_B^{-1}(t_l^B)^{-1}}. \quad (118)$$

The space $\mathcal{P}_{\leq \varepsilon_1}$ is invariant under every Y_μ . Since the a_B -eigenspace of $Y_{-\theta}^{B_n^{(1)}}$ on this space is one-dimensional, the polynomial in (118) is a common Y -eigenvector.

7.4 The corresponding polynomial of type $C_n^{(1)}$

Set

$$\eta_i := \frac{\varepsilon_i}{\sqrt{2}}, \quad \beta_0 := \eta_1 + \eta_2 = \frac{\varepsilon_1 + \varepsilon_2}{\sqrt{2}}, \quad y_i := X_{\eta_i}^t. \quad (119)$$

The variables y_i belong to an ambient Laurent polynomial ring; all monomials below lie in $\mathbb{F}[\check{Q}^t]$.

Lemma 7.3 (Triangular support of β_0). *For the $C_n^{(1)}$ -order,*

$$\{\beta \in \check{Q}^t : \beta \leq \beta_0\} = \{\beta_0, 0\}. \quad (120)$$

Proof. In the η -coordinates,

$$\check{Q}^t = \left\{ \sum_{j=1}^n m_j \eta_j : m_j \in \mathbb{Z}, \sum_{j=1}^n m_j \equiv 0 \pmod{2} \right\}. \quad (121)$$

Write

$$\beta_+ = \sum_{j=1}^n m_j \eta_j. \quad (122)$$

Since

$$(\eta_i, \eta_j) = \frac{1}{2} \delta_{ij}, \quad (123)$$

the simple coroots of the scaled C_n -system are

$$(\alpha_i^t)^\vee = 2(\eta_i - \eta_{i+1}) \quad (i < n), \quad (\alpha_n^t)^\vee = 2\eta_n. \quad (124)$$

Consequently,

$$(\beta_+, (\alpha_i^t)^\vee) = m_i - m_{i+1} \quad (i < n), \quad (\beta_+, (\alpha_n^t)^\vee) = m_n. \quad (125)$$

Thus β_+ is dominant if and only if

$$m_1 \geq m_2 \geq \cdots \geq m_n \geq 0. \quad (126)$$

Suppose that

$$\beta_+ \leq_{\text{dom}} \beta_0, \quad \beta_0 = \eta_1 + \eta_2. \quad (127)$$

Then

$$\beta_0 - \beta_+ = \sum_{i=1}^n c_i \alpha_i^l, \quad c_i \geq 0. \quad (128)$$

The sum of the η -coordinates of the right-hand side is $2c_n$. Hence

$$\sum_{i=1}^n m_i = 2 - 2c_n \leq 2. \quad (129)$$

This sum is nonnegative and even. If it is nonzero, the decreasing nonnegative coordinates m_i are therefore either

$$(1, 1, 0, \dots, 0) \quad \text{or} \quad (2, 0, \dots, 0). \quad (130)$$

The second possibility is excluded because

$$\beta_0 - 2\eta_1 = \eta_2 - \eta_1 = -\alpha_1^l \quad (131)$$

is not a nonnegative combination of simple roots. Thus

$$\beta_+ = 0 \quad \text{or} \quad \beta_+ = \beta_0. \quad (132)$$

Finally, let

$$J = \{i : s_i \beta_0 = \beta_0\}. \quad (133)$$

If w_J is the longest element of W_J , then

$$v(\beta_0) = w_0 w_J \quad (134)$$

is the unique maximal minimal representative of $\dot{W}/W_J$. The elements $v(\beta)$, with $\beta \in \dot{W}\beta_0$, run through these representatives. Therefore

$$v(\beta_0) \leq v(\beta) \quad (135)$$

forces $\beta = \beta_0$. □

Hence

$$E_{\beta_0}^{C_n^{(1)}} = X_{\beta_0}^l + c_C. \quad (136)$$

Put

$$L_C := (t_s^C)^{1/2}, \quad S_C := (t_l^C)^{1/2}. \quad (137)$$

Here L_C is the short-root parameter and S_C is the long-root parameter. Since $-2\eta_1 \in M^l$ is anti-dominant,

$$Y_{-2\eta_1}^{C_n^{(1)}} = T_1 T_2 \cdots T_{n-1} T_n T_{n-1} \cdots T_2 T_1 T_0. \quad (138)$$

In particular, $T_1, \dots, T_{n-1}$ carry L_C , whereas T_0, T_n carry S_C .

Since

$$s_0(y_1) = q_C^{-1} y_1^{-1}, \quad X_{-\alpha_0}^t = q_C y_1^2, \quad (139)$$

we have

$$T_0(y_1 y_2) = q_C^{-1} S_C^{-1} y_1^{-1} y_2. \quad (140)$$

Applying the remaining operators in (138) gives

$$\boxed{\gamma_{-2\eta_1}^{C_n^{(1)}} X_{\beta_0}^t = q_C^{-1} (t_s^C)^{-(n-2)} (t_l^C)^{-1} X_{\beta_0}^t + q_C^{-1} (t_s^C)^{n-2} (1 - t_s^C).} \quad (141)$$

The word in (138) contains $2n - 2$ short-root generators and two long-root generators. Therefore

$$\boxed{\gamma_{-2\eta_1}^{C_n^{(1)}} 1 = (t_s^C)^{n-1} t_l^C.} \quad (142)$$

Set

$$a_C := q_C^{-1} (t_s^C)^{-(n-2)} (t_l^C)^{-1}, \quad (143)$$

$$b_C := q_C^{-1} (t_s^C)^{n-2} (1 - t_s^C), \quad (144)$$

$$d_C := (t_s^C)^{n-1} t_l^C. \quad (145)$$

Then

$$\gamma_{\beta_0}^{C_n^{(1)}} (-2\eta_1) = a_C, \quad \gamma_0^{C_n^{(1)}} (-2\eta_1) = d_C. \quad (146)$$

If $a_C \neq d_C$, the eigenvalue equation yields

$$\boxed{c_C = \frac{q_C^{-1} (t_s^C)^{n-2} (t_s^C - 1)}{(t_s^C)^{n-1} t_l^C - q_C^{-1} (t_s^C)^{-(n-2)} (t_l^C)^{-1}}.} \quad (147)$$

Thus

$$\boxed{E_{\beta_0}^{C_n^{(1)}} = X_{\beta_0}^t + \frac{q_C^{-1} (t_s^C)^{n-2} (t_s^C - 1)}{(t_s^C)^{n-1} t_l^C - q_C^{-1} (t_s^C)^{-(n-2)} (t_l^C)^{-1}}.} \quad (148)$$

7.5 An explicit B_n - C_n duality identity

Take

$$\lambda = \varepsilon_1, \quad \beta_0 = \frac{\varepsilon_1 + \varepsilon_2}{\sqrt{2}}. \quad (149)$$

The spectral-point definitions give

$$X_{\beta_0}^t(\text{sp}_B(\varepsilon_1)) = a_B, \quad X_{\beta_0}^t(\text{sp}_B(0)) = d_B, \quad (150)$$

$$x_1(\text{sp}_C(\beta_0)) = a_C, \quad x_1(\text{sp}_C(0)) = d_C. \quad (151)$$

Transporting the $B_n^{(1)}$ -coefficients to the $C_n^{(1)}$ -coefficient field by J_{BC} , the normalized duality identity is Therefore

$$\frac{E_{\lambda}^R(\text{sp}_{R^t}(\beta_0))}{E_{\lambda}^R(\text{sp}_{R^t}(0))} = \frac{a_C + J_{BC}(c_B)}{d_C + J_{BC}(c_B)}, \quad (152)$$

and

$$\frac{E_{\beta_0}^{R^t}(\text{sp}_R(\lambda))}{E_{\beta_0}^{R^t}(\text{sp}_R(0))} = \frac{j_{BC}(a_B) + c_C}{j_{BC}(d_B) + c_C}. \quad (153)$$

Thus the duality becomes

$$\boxed{\frac{a_C + j_{BC}(c_B)}{d_C + j_{BC}(c_B)} = \frac{j_{BC}(a_B) + c_C}{j_{BC}(d_B) + c_C}}. \quad (154)$$

Equivalently, use the common notation

$$q := q_B = q_C, \quad t_l := t_l^B = t_s^C, \quad t_s := t_s^B = t_l^C. \quad (155)$$

Then

$$a_B = q^{-1}t_l^{-1}, \quad d_B = t_l^{2n-3}t_s, \quad (156)$$

$$a_C = q^{-1}t_l^{-(n-2)}t_s^{-1}, \quad d_C = t_l^{n-1}t_s, \quad (157)$$

and

$$c_B = \frac{q^{-1}t_l^{n-2}(t_s - 1)}{t_l^{2n-3}t_s - q^{-1}t_l^{-1}}, \quad (158)$$

$$c_C = \frac{q^{-1}t_l^{n-2}(t_l - 1)}{t_l^{n-1}t_s - q^{-1}t_l^{-(n-2)}t_s^{-1}}. \quad (159)$$

Thus (154) becomes

$$\boxed{\frac{a_C + c_B}{d_C + c_B} = \frac{a_B + c_C}{d_B + c_C}}. \quad (160)$$

Both sides are equal to

$$\boxed{\frac{q^{-1}t_l^{n-2}(t_l + t_s - 1) - q^{-2}t_l^{-(n-1)}t_s^{-1}}{t_l^{3n-4}t_s^2 - q^{-1}t_l^{n-2}}}. \quad (161)$$

This verifies evaluation-duality for $\lambda = \varepsilon_1$ and $\beta_0 = (\varepsilon_1 + \varepsilon_2)/\sqrt{2}$.

8 Guiding Principle

The X -operators are coordinates:

$$X_\beta \quad \text{acts by multiplication.} \quad (162)$$

The Y -operators are spectral operators:

$$Y_\mu \quad \text{acts by } q\text{-difference operators.} \quad (163)$$

The DAHA involution swaps them:

$$\boxed{\text{coordinates} \longleftrightarrow \text{spectral variables.}} \quad (164)$$

This gives Macdonald duality.

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